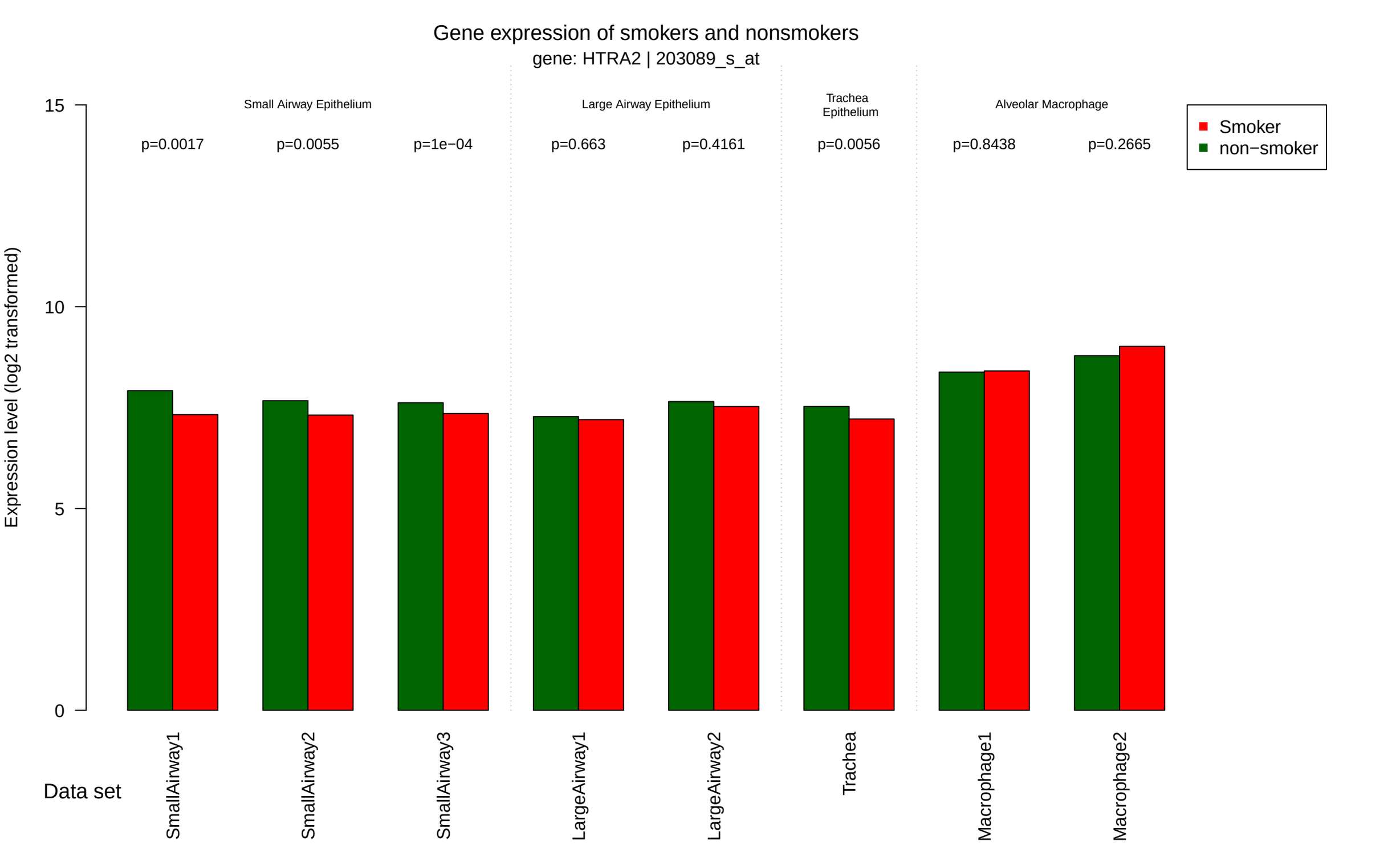
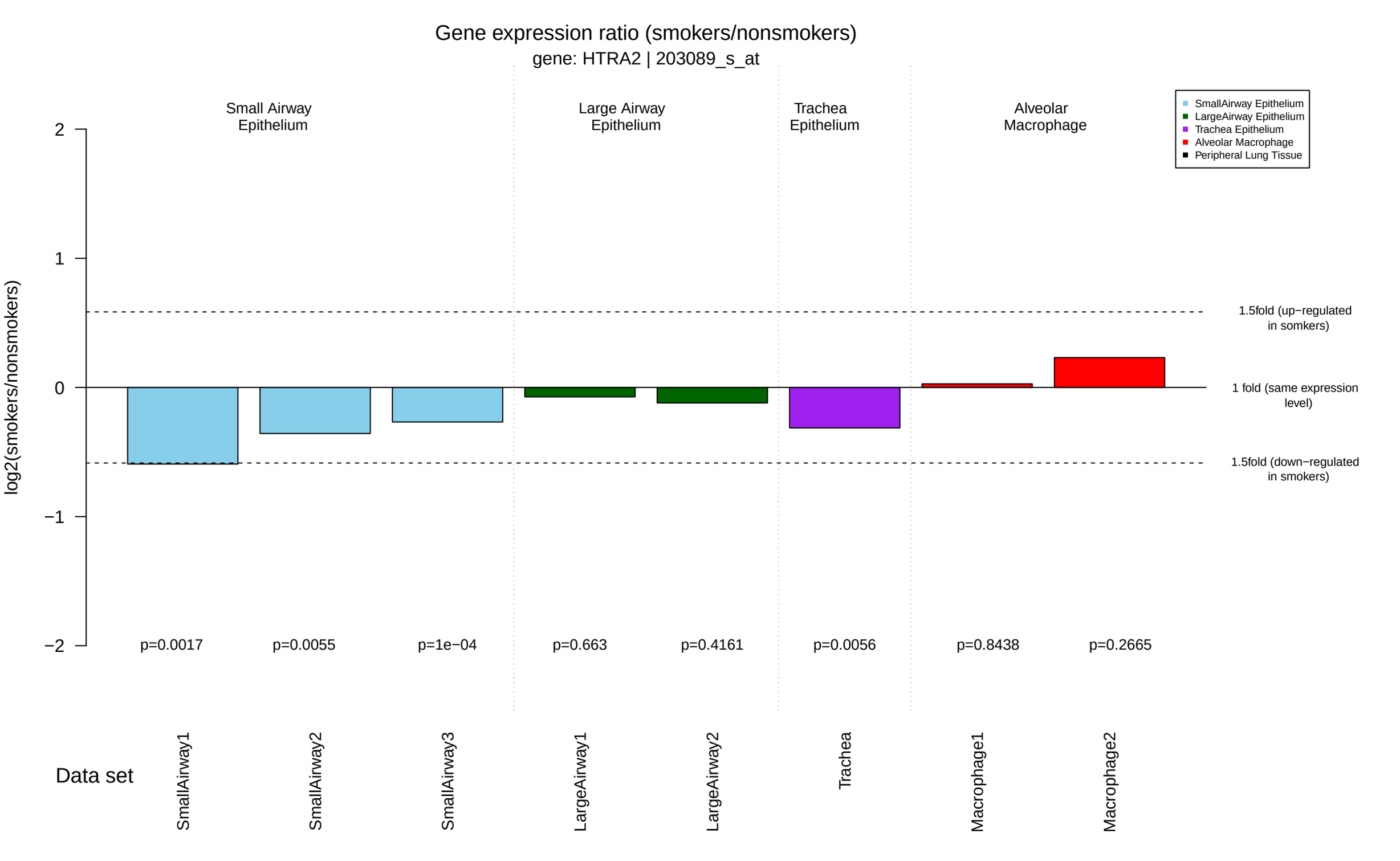
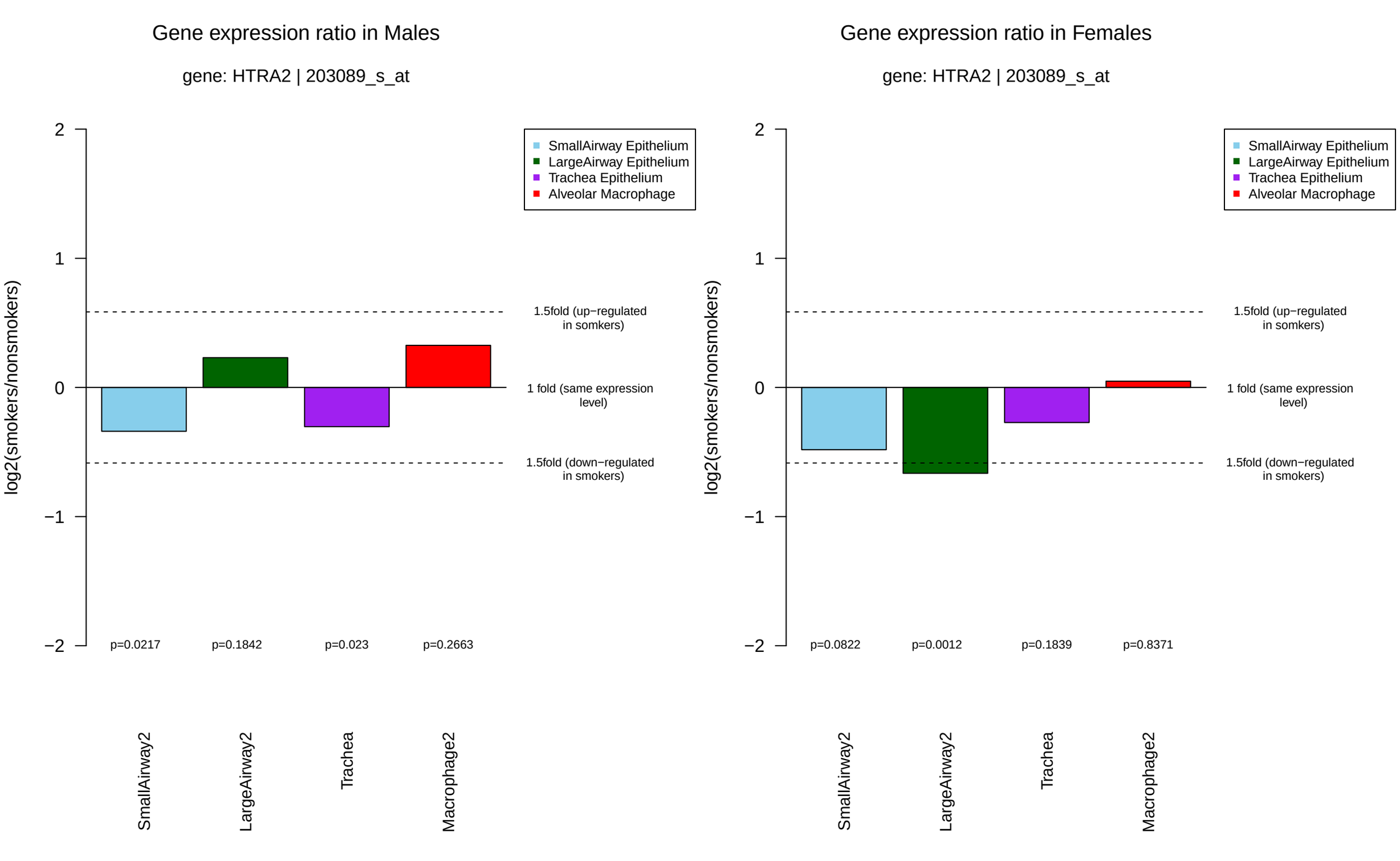
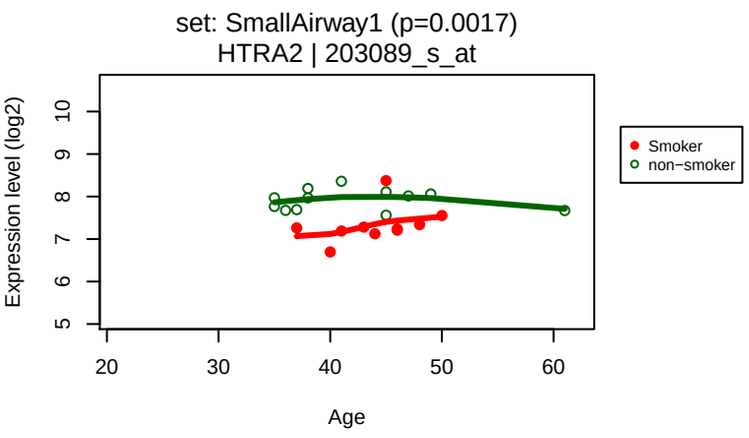
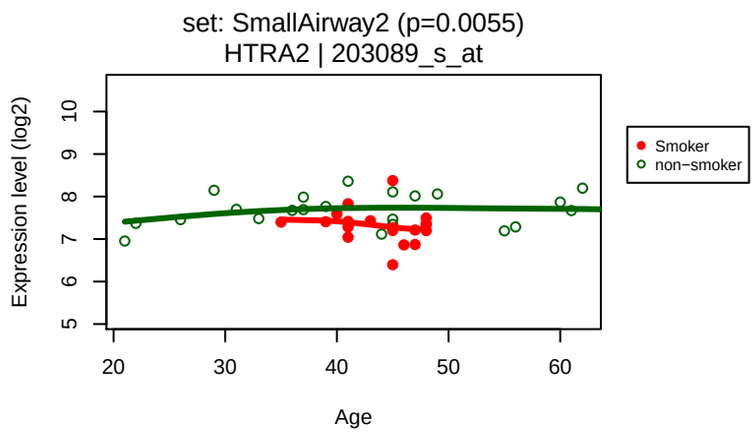
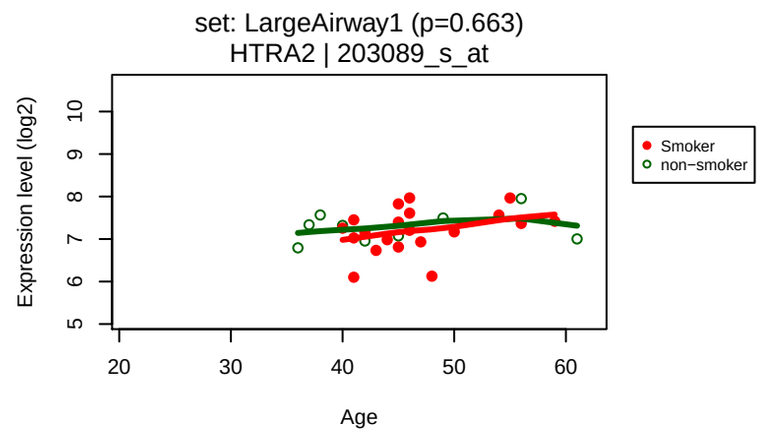

Gene expression of smokers and nonsmokers
gene: HTRA2 | 203089_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0017
p=0.0055
p=1e−04
p=0.663
p=0.4161
p=0.0056
p=0.8438
p=0.2665
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

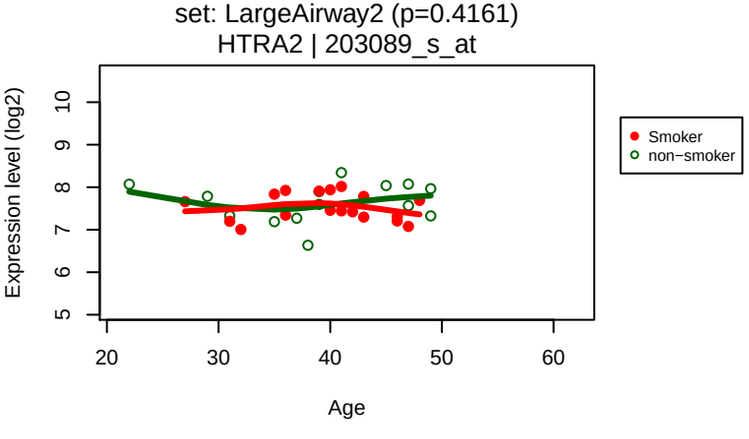

Gene expression ratio (smokers/nonsmokers)
gene: HTRA2 | 203089_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0017
p=0.0055
p=1e−04
p=0.663
p=0.4161
p=0.0056
p=0.8438
p=0.2665
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

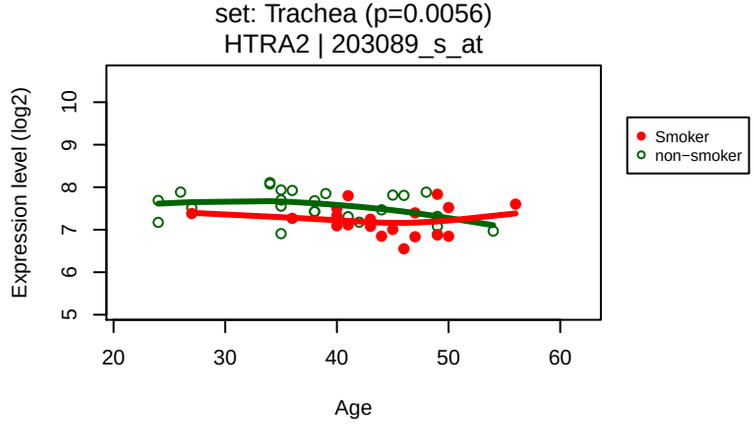

Gene expression ratio in Males
gene: HTRA2 | 203089_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0217
p=0.1842
p=0.023
p=0.2663
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HTRA2 | 203089_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0822
p=0.0012
p=0.1839
p=0.8371
SmallAirway2
LargeAirway2
Trachea
Macrophage2

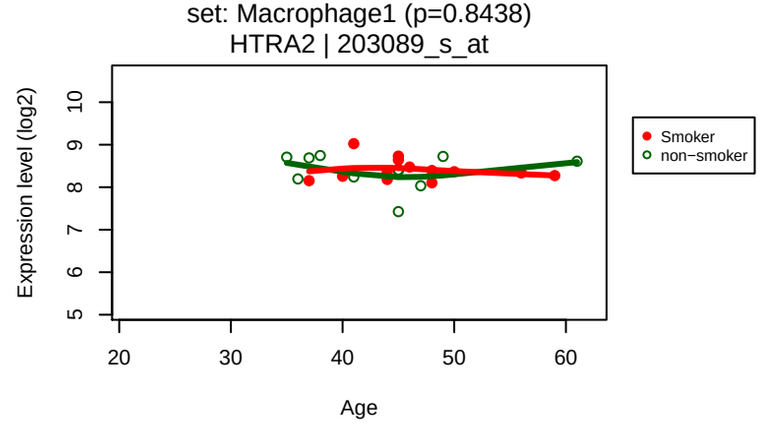

set: SmallAirway1 (p=0.0017)
HTRA2 | 203089_s_at
Expression level (log2)
Age
Smoker
non−smoker

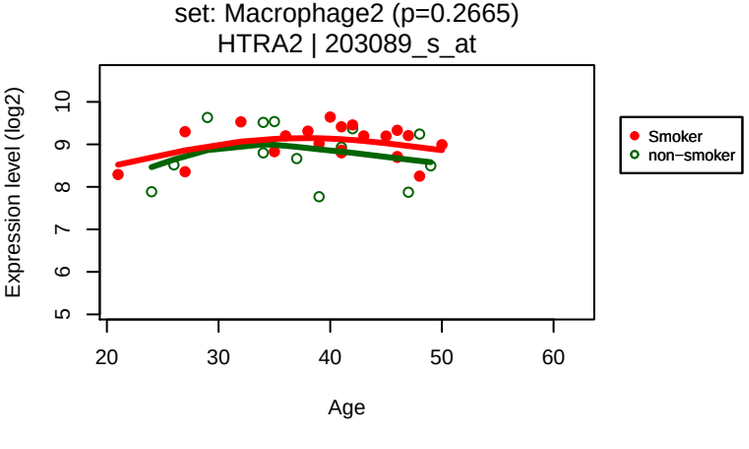

set: SmallAirway2 (p=0.0055)
HTRA2 | 203089_s_at
Expression level (log2)
Age
Smoker
non−smoker

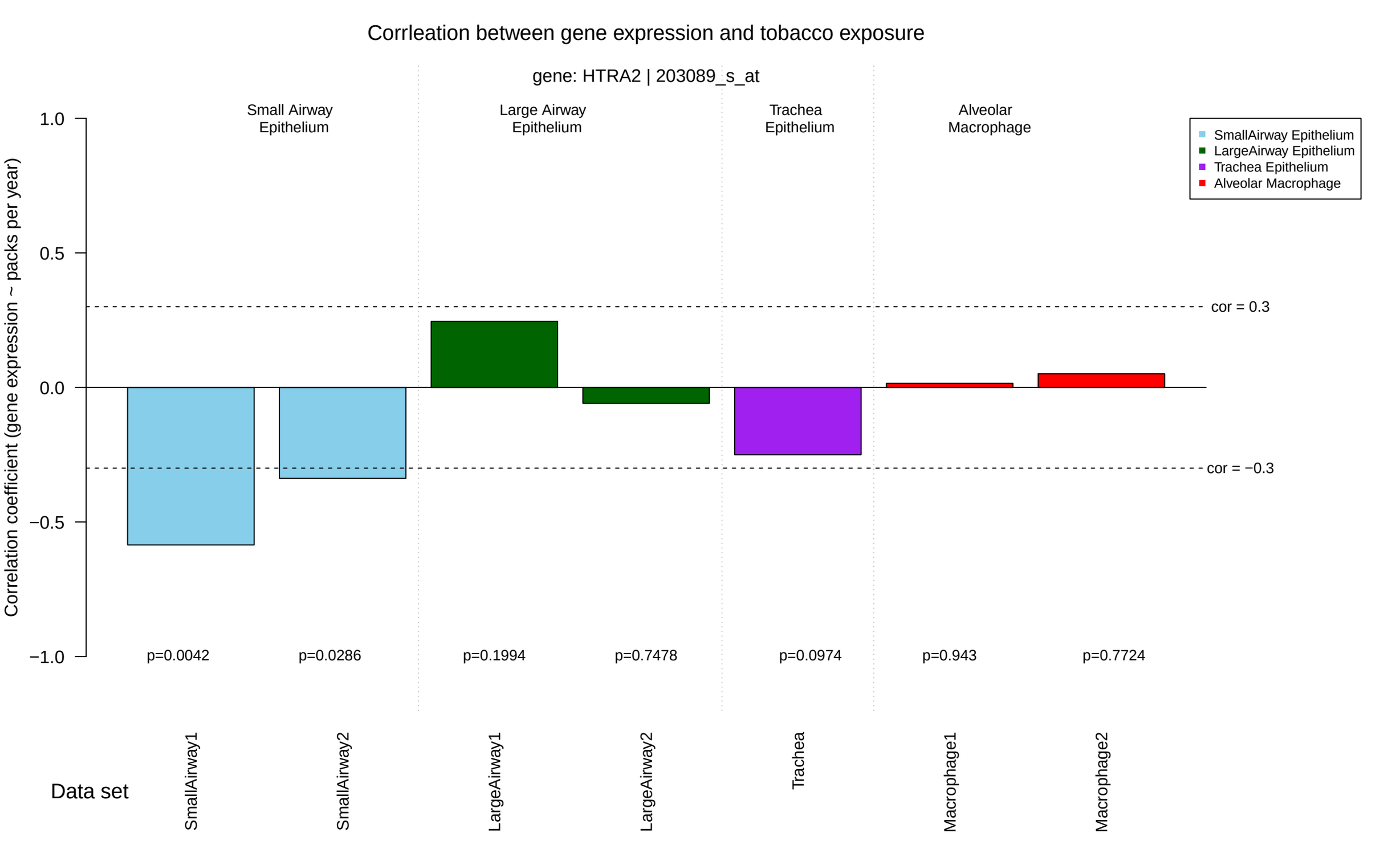

set: LargeAirway1 (p=0.663)
HTRA2 | 203089_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4161)
HTRA2 | 203089_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0056)
HTRA2 | 203089_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8438)
HTRA2 | 203089_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2665)
HTRA2 | 203089_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HTRA2 | 203089_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0042
p=0.0286
p=0.1994
p=0.7478
p=0.0974
p=0.943
p=0.7724
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2